Proof. Without loss of generality, we assume integral lengths of intervals, and later explain how to generalize to real lengths. We discretize the weight of intervals into weight units, and define a weight element w for each unit of weight. Let $W_I = \{w_1, ..., w_{w(I)}\}$ be the set of weight elements corresponding to the weight of interval I, with $W_I \cap W_J = \emptyset$ for any two distinct intervals I, J . Let $W_{opt} = \bigcup_{I \in OPT} W_I$, and $W_{alg} = \bigcup_{I \in ALG} W_I$. The sets H and H_I are defined as for the unweighted algorithms. Lastly, let $C_{opt}(I)$ be the set of optimal intervals in OPT that conflict with interval I.

We argue for the existence of an injective mapping $F : W_{opt} \rightarrow W_{alg} \cup H$ as follows: Let I_{opt} be an optimal interval. If I_{opt} is taken by the algorithm, we map the elements of $W_{I_{opt}}$ to their corresponding elements in W_{alg}. If I_{opt} is not taken by the algorithm, there are two cases. The first case is that I_{opt} did not conflict with any interval in the solution, but $Prd(I_{opt}) = 0$. In this case, we know that $\eta(I_{opt}) = |H_{I_{opt}}| = w(I_{opt})$, and we can map the weight elements of I_{opt} to error elements in $H_{I_{opt}}$.

The other case is that I_{opt} conflicted with at least one interval in the solution. Let that conflicting interval be I_c. It could also be that $Prd(I_{opt}) = 0$, and $H_{I_{opt}}$ would have error elements we can use, but we will assume the worst case of $Prd(I_{opt}) = 1$. In this case, it could be $|W_{I_{opt}}| > |H_{I_c}|$, and we cannot map the weight elements to error elements exclusively. We can, however, map the optimal weight elements, to elements in $H_{I_c} \cup W_{I_c}$. To see that there will always be sufficiently many unmapped elements, notice that $|H_{I_c} \cup W_{I_c}| \geq |\bigcup_{I \in C_{opt}(I_c)} W_I|$. This is because $|H_{I_c}| = |\bigcup_{I \in C_{opt}(I_c)} W_I| - |W_I|$, and $W_{I_c} \cap H_{I_c} = \emptyset$ always holds. We conclude that $|W_{alg}| + |H| \geq |W_{opt}|$, and we get the desired bound.

To adapt the proof to real lengths, instead of considering sets of error and weight elements, we can define a transport plan using two transport matrices H and W of size $OPT \times ALG$. H_{ij} (respectively W_{ij}) corresponds to the (real) amount of weight, or mass, mapped from $I_i \in OPT$ to the amount of error (resp. weight) introduced by $I_j \in ALG$. We can define these matrices such that for $1 \leq i \leq OPT$, $\sum_{1 \leq j \leq ALG} H_{ij} + W_{ij} = w(I_i)$, for $1 \leq j \leq ALG$, we have that $\sum_{1 \leq i \leq OPT} H_{ij} \leq \eta(I_j)$ and $\sum_{1 \leq i \leq OPT} W_{ij} \leq w(I_j)$. □

Theorem 3.6. *For every deterministic algorithm, there exist a proportional weights instance and predictions, such that $ALG = OPT - \eta$.*

Proof. Let I_1 arrive first with $Prd(I_1) = 0$. If the algorithm doesn't accept I_1, no more intervals arrive, and we have that $ALG = 0$, $OPT = w(I_1)$, and $\eta = w(I_1)$. If the algorithm accepts I_1, let two intervals I_2 and I_3 arrive next, with $w(I_2) = w(I_1)$, $Prd(I_2) = 1$, $w(I_3) = 2w(I_1)$ and $Prd(I_3) = 0$. In this case we have $ALG = w(I_1)$, $OPT = 3w(I_1)$, and $\eta = \eta(I_3) = 2w(I_1)$. In both cases the equality holds. One can repeat this construction for an asymptotic result. □

Corollary 3.7. *Algorithm Naive is optimal for proportional weights in the model of irrevocable acceptances.*

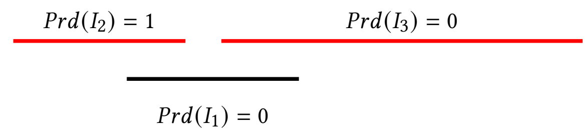

Figure 3: Instance of Theorem 3.6, with $w(I_2) = w(I_1), w(I_3) = 2w(I_1)$.

4 REVOCABLE ACCEPTANCES

Given the difficulty of the problem(s) in the conventional online model, we now consider the case where acceptances are revocable, but rejections are final, a relaxation of the model that is commonly studied for the problem of interval selection. A new interval can now always be accepted by displacing any intervals in the solution conflicting with it. For unit weights, Borodin and Karavasilis [12] give an optimal algorithm that is $2k$-competitive, where k is the number of distinct interval lengths. We will refer to this algorithm of [12] as the *BK2K* algorithm. *BK2K* is a greedy algorithm, always accepting a new interval when there is no conflict, and whenever a conflict exists, the new interval is accepted only if it is properly included in an interval currently in the solution. We use that as the base logic for our predictions algorithm, and add one more replacement rule, which accepts a new interval I that is only involved in partial conflicts, if $Prd(I) = 1$. Furthermore, an interval accepted by that rule gets marked, to make sure it cannot be replaced by that rule again. We call this algorithm `Revoke-Unit` 2. Interestingly, this rule of locally *following the predictions once*, suffices to give us 1-consistency.

Algorithm 2 `Revoke-Unit`

$M \leftarrow \emptyset$ ▷ Set of marked intervals
$S \leftarrow \emptyset$ ▷ Solution set
On the arrival of I:
$I_s \leftarrow$ Set of intervals currently in the solution conflicting with I
if $I_s = \emptyset$ or ($I_s = \{I'\}$ and $I \subset I'$) **then**
  if $I' \in M$ **then**
    $M \leftarrow M \cup \{I\}$
  $S \leftarrow S \cup \{I\} \setminus \{I'\}$ ▷ Take I and discard I' if necessary
else if I is only involved in partial conflicts **and** $Prd(I) = 1$ **and** $I_s \cap M = \emptyset$ **then**
  $S \leftarrow S \cup \{I\} \setminus I_s$ ▷ Take I and discard conflicting intervals
  $M \leftarrow M \cup \{I\}$

Theorem 4.1. *Algorithm 2 achieves $ALG \geq OPT - \eta$.*

Proof. We follow the same approach as in the proof of Theorem 3.1, mapping optimal intervals to intervals taken by the algorithm, and to error. The main difference is that because of revoking, this mapping might be redefined throughout the execution of the algorithm. As before, we let H be the set of error elements, and $H_I \subseteq H$ be the set of error elements introduced by $\eta(I)$. Let I_{opt} be an optimal interval. We will define an injective mapping $F : OPT \rightarrow ALG \cup H$ as follows: If I_{opt} is taken by the algorithm, it is initially mapped onto itself. If I_{opt} is later replaced, it must be because of a partial conflict (w.l.o.g. no interval is subsumed by an

optimal interval) with a new interval I' with $Prd(I') = 1$. In this case, I_{opt} will be mapped to an error element in $H_{I'}$, or if no further optimal intervals that conflict with I are yet to arrive, it will be mapped to I. In both, subcases it will never be remapped.
Consider now the case of I_{opt} being rejected upon arrival. This can only happen if it is involved in (at most two) partial conflicts. There are two possible cases. The first case is that $Prd(I_{opt}) = 0$, and therefore $|H_{I_{opt}}| = 1$, in which case we map I_{opt} to the error element of its own prediction. The second case is that $Prd(I_{opt}) = 1$, but at least one of the conflicting intervals was marked. Let I_c be one of the marked, partially conflicting intervals. If I_c was marked by being taken through a partial-conflict replacement, it means that $Prd(I_c) = 1$, and $|H_{I_c} \cup \{I_c\}| > 0$, in which case we can map I_{opt} to an element $h_c \in H_{I_c} \cup \{I_c\}$.
If I_c was instead marked by a proper-inclusion-replacement, we trace the original interval that got marked through a partial-conflict-replacement. Call that interval $I_c^{'}$. It holds that $I_c^{'}$ conflicts with I_c, and therefore also conflicts with I_{opt}. Moreover, for $I_c^{'}$ to have been accepted, it must be that $Prd(I_c^{'}) = 1$ and $|H_{I_c^{'}} \cup \{I_c^{'}\}| > 0$. In this case, we map I_{opt} to an element $h_{c'} \in H_{I_c^{'}} \cup \{I_c^{'}\}$. In conclusion, we have that $ALG + |H| \geq OPT$, and we get the desired bound. ◻

We note that the performance of algorithm 2 on the instance of Theorem 3.3 is exactly equal to $OPT - \eta$, and we get the following lemma.

Lemma 4.2. *The performance of algorithm 2 cannot be better than $OPT - \eta$.*

We next show that the robustness of algorithm `Revoke-Unit` nearly matches the optimal online guarantee.

Theorem 4.3. *With at most k distinct interval lengths, algorithm 2 is $(2k + 1)$-robust.*

Proof. We use a charging argument and show that an interval taken by the algorithm can be charged by at most $2k + 1$ optimal intervals. As soon as an optimal interval arrives, we map it to an interval already taken by the algorithm or itself. When an interval is replaced during the execution, all optimal intervals charged to it up to that point, will now be charged to the new interval that was accepted. We build upon the proof of Theorem 3.2 in [12]. In the case of the *BK2K* algorithm ([12]), it is true that for every predecessor trace $\mathcal{P}$, and consecutive intervals $(I_i, I_{i+1}) \in \mathcal{P}, \Phi(I_{i+1}) \leq \Phi(I_i)+2$, and the length of every predecessor trace is at most k. While the former is still true for algorithm `Revoke-Unit`, the latter is not, and that is because we have an additional replacement rule. However, we argue that for every $I_i \in \mathcal{P}$, if $I_j,\ j > i$ is the next interval in the trace that was accepted through proper-inclusion, it is true that $TC(I_j) \leq TC(I_i) + 3$.

Figure 4 shows how to maximize charge on the event of a partial-conflict replacement. Before being replaced by a partially-conflicting interval, I_1 can be directly charged by at most two optimal intervals (I_{opt}^1, I_{opt}^2), one on each side. After I_2 replaces I_1, it can also be directly charged by two optimal intervals, but only if I_{opt}^2 was not charged to I_1 earlier. In other words, if I_2 is directly charged by two new intervals, it means that I_1 was directly charged by at most one, concluding that $\Phi(I_2) \leq TC(I_1) + 3$.

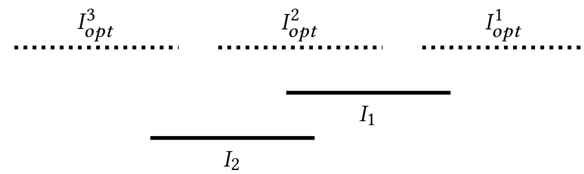

Figure 4: Maximum charge through partial-conflict replacement.

Finally, notice that because the mark of an interval carries over when it is replaced, the event of a partial-conflict replacement can occur at most once in each predecessor trace, and excluding at most one subsequence $(I_i, I_r, I_j) \in \mathcal{P}$ where $TC(I_j) \leq TC(I_i)+3$, it holds that for $(I_b, I_{b+1}) \in \mathcal{P}$, $\Phi(I_{b+1}) \leq \Phi(I_b) + 2$, giving us a worst case competitive ratio of $2k + 1$. ◻

Corollary 4.4. *With at most k distinct interval lengths, and predictions with total error η, Algorithm Revoke-Unit achieves $ALG \geq \max\{OPT - \eta, \frac{OPT}{2k+1}\}$.*

Notice how we can choose not to carry over the mark when proper-inclusion replacement occurs, and get a $3k$-robust algorithm. Such an algorithm is prone to follow the prediction more often, and it can outperform `Revoke-Unit` for some small values of error caused by adversarial predictions.

We now look at the case of proportional weights. In the conventional online setting, Garay et al. [19] give a $2\phi + 1 \approx 4.236-$competitive algorithm, while Tomkins [32] gives a matching lower bound. They call their optimal algorithm `LR` (for *length of route*), and we include it here for completeness. Unlike the case of unit weights, we now want to accept intervals that occupy as much of the line as possible. Algorithm `LR` works greedily by always accepting a new interval with no conflicts, and when there are conflicts, it accepts the new interval if its length is at least ϕ times greater than the largest conflicting interval. More generally, using parameter $\beta \geq \phi$, we have the following lemma:

Lemma 4.5 (Garay et al. [19]). *Algorithm `LR` with parameter $\beta \geq \phi$ is $(2\beta + 1)$-competitive for the problem of interval selection with proportional weights.*

Algorithm 3 LR [19]

```
  Parameter β = φ                         ▷ optimal value for parameter β
  On the arrival of I:
  I_s ← Set of intervals currently in the solution conflicting with I
  if  w(I) > β · max{w(J) : J ∈ I_s} then
      Accept I and displace conflicts
      Return
```

Instead of using algorithm `LR` as the base of our predictions algorithm, we consider a slightly modified version, which we refer to as LR′, and which compares the weight of the new interval to the sum of the weights of the conflicting intervals, instead of looking only at the longest interval. Although we do not know the exact performance of algorithm LR′ in the online model, we conjecture it is also $(2\phi + 1)$-competitive.

In trying to utilize predictions in the case of proportional weights, we first make the following observations:

Observation 4.6. *1-consistency is unattainable while maintaining bounded robustness.*

Proof. To be 1-consistent, the algorithm must be able to replace an interval with an arbitrarily smaller one that is part of the optimal solution. The adversary could then stop the instance, forcing arbitrarily bad robustness. □

Observation 4.7. *In order to have bounded robustness, it must be that a new interval that is sufficiently large (small[2]) must always be accepted (rejected).*

Definition 4.8 ($\alpha-$increasing). An α-increasing algorithm never accepts a new conflicting interval that is less than α times the longest interval it conflicts with.

Lemma 4.9. *An α-increasing algorithm (greedy or non-greedy), cannot be better than $(2\alpha + 1)$-consistent.*

Proof. Let an interval I_1 arrive first. Let I_2 and I_3 be intervals that partially conflict with I_1 on either side, and $w(I_2) = w(I_3) = \alpha \cdot w(I_1) - \epsilon$. Let I_4 with $w(I_4) = w(I_1) - 2\epsilon$ be an interval that is fully subsumed by I_4. This instance is depicted in figure 5. The algorithm will never replace I_1, while the optimal solution is made of $\{I_2, I_3, I_4\}$. □

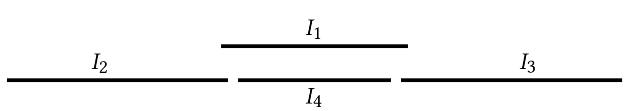

Figure 5: Consistency bound for α-increasing algorithms.

Algorithm LR is a ϕ-increasing algorithm, while the algorithm by Woeginger [33] for the real-time model is 2-increasing. Our predictions algorithm 4 `Revoke-Proportional` is 1-increasing. The algorithm works like LR′, with one additional replacement rule that accepts a new interval that is predicted to be optimal, even if it is not sufficiently larger than what it conflicts with. More precisely, if a new interval is predicted to be optimal and is at least as big as the sum of the weights of the intervals it conflicts with, and none of the conflicting intervals were predicted to be optimal, it will be accepted through the predictions rule. The algorithm takes a parameter $\lambda > 1$, which can be thought of as an indicator of how much the predictions are trusted. As λ increases, the consistency bound improves.

Theorem 4.10. *Algorithm Revoke-Proportional is $\frac{3\lambda}{\lambda-1}$-consistent.*

[2]More accurately, an interval reducing ALG sufficiently much.

Algorithm 4 `Revoke-Proportional` Parameter: $\lambda > 1$

On the arrival of I:
$I_s \leftarrow$ Set of intervals currently in the solution conflicting with I
Let $w_c = \sum_{J \in I_s} w(J)$ ▷ Total weight of conflicting intervals
if $w(I) \geq \lambda \cdot w_c$ **then** ▷ Main replacement rule
  Accept I and displace conflicts
  Return
else if $Prd(I) = 1$ **then** ▷ Predictions rule
  if $(w(I) \geq w_c$ and $|\{J : J \in I_s$ and $Prd(J) = 1\}| = \emptyset)$ **then**
    Accept I and displace conflicts
    Return

Proof. We consider the optimal solution OPT consistent with the fully accurate predictions. We will show that throughout the execution of the algorithm, we have that $\Phi(I) \leq \mu \cdot w(I)$, for every I in the current solution. In the end, we have that $\sum_{I \in ALG} \Phi(I) = OPT$, giving us the μ-consistency of the algorithm.

As in the proof of Theorem 4.3, we consider the notions of *transfer charge* (TC), and *direct charge* (DC). We can express $\Phi(I) = TC(I) + DC(I)$. A transfer charge occurs whenever accepting a new interval I replaces intervals currently in the solution. In that case, the total charge of those conflicting intervals is passed on as transfer charge to I. Any additional charge to I after its acceptance is through direct charge, namely rejection of subsequent optimal intervals conflicting with I. We will write $DC_J(I)$ to denote the amount of direct charge from interval J to interval I. Whenever an optimal interval is accepted, we consider its weight being directly charged to itself, and it cannot be directly charged again.

Whenever an optimal interval is rejected upon arrival, we charge its weight to the intervals it conflicts with, with its weight being distributed to all its conflicting intervals, in proportion to their weight. Specifically, let I_o be the newly arrived optimal interval that is rejected, and I_s denote the set of conflicting intervals. Each interval $J \in I_s$ is directly charged $DC_{I_o}(J) = w(I_o)\frac{w(J)}{w_c} \leq w(I_o)$. Furthermore, for an optimal interval to have been rejected, it must be that even the predictions rule failed, and because the predictions are accurate, it must have failed because $w(I_o) < w_c$. Because of this, we get that $w(I_o)\frac{w(J)}{w_c} \leq w(J)$, and therefore $DC_{I_o}(J) \leq \min\{w(I_o), w(J)\}$. An interval $I \in ALG$ can be directly charged by at most three different types of optimal intervals: 1) smaller intervals that are subsumed by it, 2) an optimal interval partially conflicting on the left, and 3) an optimal interval partially conflicting on the right. In the case of smaller optimal intervals subsumed by I, the total amount of direct charge from those intervals can be at most $w(I)$. Given that each of the two possible partially conflicting intervals can directly charge I at most $w(I)$, we conclude that for every $I \in ALG$:

$$DC(I) \leq 3w(I) \tag{1}$$

We omitted the case where the rejected optimal interval subsumes I, because in that case $DC(I) = w(I)$ and 1 holds trivially. We now focus on the total amount of charge on any interval $I \in ALG$. Let:

$$\mu = \frac{3\lambda}{\lambda - 1}$$